CardDemo (COBOL) — Architecture and Interfaces

Derived from the COBOL/CICS/VSAM/JCL source code of the original application

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1 Introduction

1.1 Purpose

This document describes the architecture and the interfaces of the original **CardDemo** mainframe application: its subsystems, the CICS transactions and programs, the screens, the communication area that the online programs pass to each other, the batch job streams, the interfaces of the batch programs, and the data sets and record layouts they share.

It is a companion to the *Software Requirements Specification* (`requirements.pdf`), which describes *what* the system does; this document describes *how it is built*. The Rust translation has its own document (`architecture-rust.pdf`).

1.2 How this document was made: derived from the code

This document was derived from the source code; it is not based on separate design documents. No architecture or interface documentation was supplied with the original application. Every statement here was obtained by reading the source in `cobol/app/` — COBOL programs (`cbl/`), copybooks (`cpy/`), BMS map definitions (`bms/`), JCL (`jcl/`), the CICS resource definitions (`csd/CARDDEMO.CSD`) and the assembler routines (`asm/`) — partly with small extraction scripts (CICS commands, `SELECT ... ASSIGN` clauses, `DEFINE CLUSTER` statements, `EXEC PGM=` steps), and checked against the code. The source was taken from the repository at commit `49ee1a5` (7 November 2026); the `cobol/` folder there is AWS CardDemo as forked from `aws-samples/aws-mainframe-modernization-carddemo` at commit `59cc6c2`, unchanged.

Consequences for the reader:

- **Where this document and the code disagree, the code is right.** The document records the code as it is, including oddities and defects (the requirements list the known defects as *KD-nn*); it does not say how the system was *meant* to work.
- Names (programs, transactions, files, fields, paragraphs) are quoted exactly as they appear in the code.
- Statements about run-time behaviour (for example navigation between screens) are read from the program logic; they were confirmed by running the original programs with GnuCOBOL on a CICS stand-in (see the Rust architecture document), not on a mainframe.

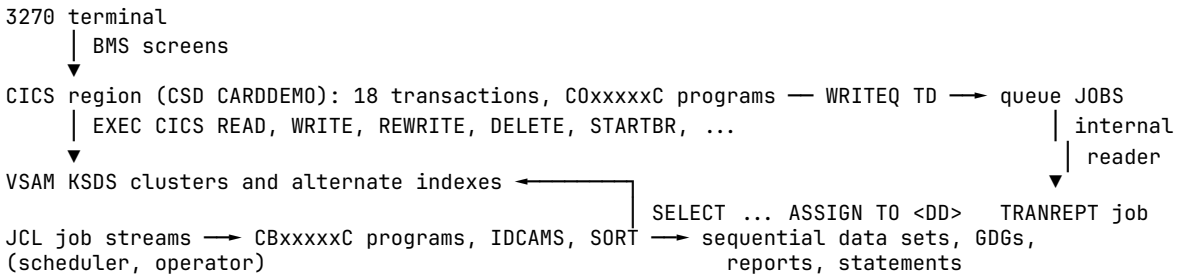
1.3 Scope

The **base application**: 17 online CICS programs, 13 batch COBOL programs, the wait utility `COBSWAIT`, 2 assembler routines, 17 BMS mapsets, 30 copybooks, 38 JCL members and the CSD. The optional modules in `cobol/app/app-*` (credit-card authorisations with IMS/DB2/MQ, transaction-type maintenance with DB2, MQ account and date inquiry) are outside the scope, as in the requirements; where base code refers to them, this document says so.

2 System overview

CardDemo is a credit-card management system with two subsystems sharing one set of VSAM files:

Subsystem	Runs under	Made of	Users
Online	CICS, 3270 terminals	17 pseudo-conversational COBOL programs with BMS maps	Customer-service users (type `U`) and administrators (type `A`)
Batch	z/OS JES, JCL	Job streams of COBOL programs, IDCAMS, DFSORT, IEBGENER, IEFBR14	Operations (scheduled, for example nightly)
Data	VSAM, QSAM	KSDS clusters with alternate indexes; sequential files and GDGs	Both



The two subsystems meet in two places only: the shared VSAM files (the online programs read and update the same master files that batch posts to), and the transient-data queue `JOBS`, through which the online report screen (`CORPT00C`) submits the transaction-report job to the internal reader.

3 Online subsystem

3.1 Programming model

Every online program is **pseudo-conversational**: a task receives a screen, processes it, sends the next screen and ends with `EXEC CICS RETURN TRANSID(<own transaction>) COMMAREA(...)`. The next key the user presses starts a new task of the same transaction, which finds its state in the communication area.

- **First entry vs. re-entry.** `EIBCALEN = 0` means the transaction was started without a communication area (directly from the terminal); programs then sign the user on again or show their first screen. Within the application, `CDEMO-PGM-CONTEXT` distinguishes the first entry into a program after an `XCTL` (`CDEMO-PGM-ENTER`, 0) from a re-entry with a screen to receive (`CDEMO-PGM-REENTER`, 1).
- **Navigation** is by `EXEC CICS XCTL PROGRAM(...) COMMAREA(...)`: the calling program sets `CDEMO-FROM-PROGRAM`/`CDEMO-FROM-TRANID` and the target in `CDEMO-TO-PROGRAM`/`CDEMO-TO-TRANID`. **PF3** returns to the program in `CDEMO-FROM-PROGRAM` (or to the menu, or to sign-on).
- **Keys** are read from `EIBAID` and compared with the `DFHAID` constants (Enter, Clear, PF1–PF24). The card and account programs map PF13–PF24 to PF1–PF12 (`YYYY-STORE-PFKEY`, copybook `CSSTRPFY`).
- **Error handling** is either `RESP(...)` on each file command with an `EVALUATE` of the response, or `EXEC CICS HANDLE ABEND LABEL(...)` with an abend routine (`COACTUPC`, `COACTVWC`, `COCRDSLC`, `COCRDUPC`).
- **Updates** run in the task's unit of work; `COACTUPC` and `COCRDUPC` take an explicit `SYNCPOINT`, and read-for-update plus a compare of old and new values guards against changes by other users.

3.2 Transactions and programs

From the CSD (`DEFINE TRANSACTION`) and the programs' `LIT-THISTRANID`/`WS-TRANID` constants:

Transaction	Program	Mapset / map	Function	Files (CICS commands)
`CC00`	`COSGN00C`	`COSGN00` / `COSGN0A`	Sign-on	`USRSEC` (READ)
`CM00`	`COMEN01C`	`COMEN01` / `COMEN1A`	Main menu (user)	— (`INQUIRE PROGRAM` for option 11)
`CA00`	`COADM01C`	`COADM01` / `COADM1A`	Admin menu	—
`CAVW`	`COACTVWC`	`COACTVW` / `CACTVWA`	Account view	`CXACAIX`, `ACCTDAT`, `CUSTDAT` (READ)
`CAUP`	`COACTUPC`	`COACTUP` / `CACTUPA`	Account update	`CXACAIX`, `ACCTDAT`, `CUSTDAT` (READ, READ UPDATE, REWRITE)

Transaction	Program	Mapset / map	Function	Files (CICS commands)
CCLI	COCRDLIC	COCRDLI / CCRDLIA	Credit card list	CARDDAT (STARTBR, READNEXT, READPREV, ENDBR)
CCDL	COCRDSLC	COCRDSL / CCRDSLA	Credit card view	CARDDAT (READ)
CCUP	COCRDUPC	COCRDUP / CCRDUPA	Credit card update	CARDDAT (READ, READ UPDATE, REWRITE)
CT00	COTRN00C	COTRN00 / COTRN0A	Transaction list	TRANSACT (browse)
CT01	COTRN01C	COTRN01 / COTRN1A	Transaction view	TRANSACT (READ)
CT02	COTRN02C	COTRN02 / COTRN2A	Transaction add	CXACAIX, CCXREF, ACCTDAT (READ), TRANSACT (browse, WRITE)
CR00	CORPT00C	CORPT00 / CORPT0A	Transaction reports	TD queue JOBS (WRITEQ TD)
CB00	COBIL00C	COBIL00 / COBIL0A	Bill payment	CXACAIX (READ), ACCTDAT (READ UPDATE, REWRITE), TRANSACT (browse, WRITE)
CU00	COUSR00C	COUSR00 / COUSR0A	User list	USRSEC (browse)
CU01	COUSR01C	COUSR01 / COUSR1A	User add	USRSEC (WRITE)
CU02	COUSR02C	COUSR02 / COUSR2A	User update	USRSEC (READ UPDATE, REWRITE)
CU03	COUSR03C	COUSR03 / COUSR3A	User delete	USRSEC (READ UPDATE, DELETE)
CDV1	COCRDSEC	—	“Developer transaction” in the CSD	The program is **not** in the source: the transaction cannot run

CICS commands used across the online programs: SEND MAP, SEND TEXT, RECEIVE MAP, RETURN, XCTL, READ, WRITE, REWRITE, DELETE, STARTBR, READNEXT, READPREV, ENDBR, SYNCPOINT, WRITEQ TD, HANDLE ABEND, HANDLE CONDITION, ABEND, INQUIRE PROGRAM, ASKTIME, FORMATTIME, ASSIGN.

3.3 Navigation

```
CC00 COSGN00C  sign-on ──(user type U)──► CM00 COMEN01C main menu ──► options 1-10 (and 11)
                       └─(user type A)──► CA00 COADM01C admin menu ──► options 1-4 (and 5-6)
F3 on a menu ──► back to sign-on;  F3 on sign-on ──► "Thank you" text, the session ends
```

The menus are table-driven. Each option is an entry (number, name, program, user type) in copybook COMEN02Y (main menu) or COADM02Y (admin menu); the menu XCTLs to the program of the chosen entry.

Main menu (COMEN02Y)	Program	Admin menu (COADM02Y)	Program
1 Account View	COACTVWC	1 User List (Security)	COUSR00C
2 Account Update	COACTUPC	2 User Add (Security)	COUSR01C
3 Credit Card List	COCRDLIC	3 User Update (Security)	COUSR02C
4 Credit Card View	COCRDSLC	4 User Delete (Security)	COUSR03C
5 Credit Card Update	COCRDUPC	5 Transaction Type List/Update (Db2)	COTRTLIC (optional module)
6 Transaction List	COTRN00C	6 Transaction Type Maintenance (Db2)	COTRTUPC (optional module)
7 Transaction View	COTRN01C		
8 Transaction Add	COTRN02C		
9 Transaction Reports	CORPT00C		
10 Bill Payment	COBIL00C		
11 Pending Authorization View	COPAUS0C (optional module)		

COMEN01C checks with EXEC CICS INQUIRE PROGRAM(...) NOHANDLE that the chosen program is installed; for the optional modules it is not, and the menu shows a message instead of transferring.

Inside the functions, the programs transfer among themselves:

- the card list COCRDLIC → card view COCRDSLC (S) or card update COCRDUPC (U) for a selected card;
- the transaction list COTRN00C → transaction view COTRN01C for a selected transaction;
- the user list COUSR00C → user update COUSR02C (U) or user delete COUSR03C (D);
- every function → back to its caller (PF3), the menu, or sign-on when there is no communication area.

3.4 The communication area (COCOM01Y)

All online programs share the 160-byte CARDDEMO-COMMAREA. It is the only interface between them: everything a program needs to know about the session travels in it.

Offset	Length	Field	Content
0	4	CDEMO-FROM-TRANID	Transaction of the calling program
4	8	CDEMO-FROM-PROGRAM	Calling program (target of PF3)
12	4	CDEMO-TO-TRANID	Transaction of the target
16	8	CDEMO-TO-PROGRAM	Target program
24	8	CDEMO-USER-ID	Signed-on user
32	1	CDEMO-USER-TYPE	A administrator, U user
33	1	CDEMO-PGM-CONTEXT	0 first entry, 1 re-entry
34	9	CDEMO-CUST-ID	Selected customer
43	25×3	CDEMO-CUST-FNAME, -MNAME, -LNAME	Customer name
118	11	CDEMO-ACCT-ID	Selected account

Offset	Length	Field	Content
129	1	CDEMO-ACCT-STATUS	Account status
130	16	CDEMO-CARD-NUM	Selected card
146	7	CDEMO-LAST-MAP	Last map sent
153	7	CDEMO-LAST-MAPSET	Last mapset sent

Several programs append a **program-specific area** after these 160 bytes (list paging keys, selected rows, the old values of an update, a confirmation flag) and pass the longer area on RETURN. A program reads it back on re-entry. The account and card programs (COACTVWC, COACTUPC, COCRDLIC, COCRDSLC, COCRDUPC) copy their area from beyond EIBCALEN even when the caller passed only the 160 bytes (KD-16 in the requirements).

3.5 Screens (BMS)

Each program has one mapset with one map (DFHMSD/DFHMDI/DFHMDF in bms/), 24 × 80 characters. The programs use the generated symbolic maps (copybooks named after the mapset, for example COSGN00 with COSGN0AI/COSGN0AO): for each field the length (L), attribute (A), colour (C) and data (I/O). Common screen content comes from copybooks: the title lines (COTTL01Y), the date and time (CSDAT01Y), messages (CSMSG01Y, CSMSG02Y), and attribute settings (CSSETATY).

3.6 CICS resources (CARDDEMO.CSD)

Resource	Name	Definition
Files	ACCTDAT, CARDDAT, CUSTDAT, CCXREF, TRANSACT, USRSEC	The KSDS clusters (see *Data*), all with READ, ADD, UPDATE, DELETE and BROWSE
Files (paths)	CARDAIX, CXACAIX	The alternate-index paths over the card file and the card cross reference
TD queue	JOBS	Extrapartition queue to the JES internal reader
Programs, mapsets	18 programs, 17 mapsets	One per transaction program and screen (COCRDSEC defined but not supplied)

The files are opened and closed by the batch jobs OPENFIL and CLOSEFIL (SDSF commands to CICS) around the batch steps that load or replace the clusters.

4 Batch subsystem

4.1 Job streams

The JCL members in jcl/ fall into four groups. Steps and programs are taken from the EXEC statements.

Group	Job	Steps (programs)	Purpose
Loading the data	ACCTFILE, CARDFILE, CUSTFILE, XREFFILE, TRANFILE, DISCGRP, TCATBALF, TRANCATG, TRANTYPE, DUSRSECJ	IDCAMS DELETE/DEFINE CLUSTER/REPRO (CARDFILE, XREFFILE, TRANFILE also DEFINE AIX, BLDINDEX, DEFINE PATH); DUSRSECJ: IEBGENER of in-stream user records, then REPRO	Create a cluster and load it from the sample data set AWS.M2.CARDDEMO.*.PS
Nightly processing	TRANBKP	REPROC proc (REPRO to the TRANSACT.BKUP GDG), IDCAMS delete/define	Back up the transaction master and define it empty
	POSTTRAN	CBTRN02C	Post the daily transactions; rejects to the DALYREJS GDG
	INTCALC	CBACT04C (PARM run date)	Interest and fees per account; new transactions to the SYSTRAN GDG
	COMBTRAN	SORT, IDCAMS REPRO	Merge the backup with the system transactions into TRANSACT.COMBINED, reload the master
	CREASTMT	IDCAMS, SORT, IDCAMS, IEFBR14, CBSTM03A (calls CBSTM03B)	Customer statements as text (STATEMNT.PS) and HTML (STATEMNT.HTML)
	TRANREPT	REPROC, SORT (INCLUDE on processing date), CBTRN03C	The transaction report for a date range (DATEPARM)
Utilities and listings	READACCT, READCARD, READCUST, READXREF	CBACT01C, CBACT02C, CBCUS01C, CBACT03C	List a master file; CBACT01C also writes three demonstration formats
	CBEXPORT, CBIMPORT	CBEXPORT, CBIMPORT	Export all masters into one multi-record file and import it back into sequential files
	WAITSTEP	COBSWAIT (calls MVSWAIT)	Wait for the time given in the PARM
	TRANIDX, DEFGDGB, DEFGDGD, REPTFILE, DALYREJS, DEFCUST, PRTCATBL, ESDSRRDS	IDCAMS, IEBGENER, SORT	Define the transaction AIX, the GDG bases and further data sets; print the category balances; ESDS/RRDS samples
Operations	OPENFIL, CLOSEFIL	SDSF	Open and close the CICS files
	CBADMCDJ	DFHCSDUP	Define the CICS resources (the CSD)

FTPJCL, INTRDRJ1, INTRDRJ2, TXT2PDF1	FTP, IDCAMS, IEBGENER, IKJEFT1B (REXX)	Transfer, internal-reader and text-to-PDF examples

CBTRN01C (verify the daily transactions against the cross reference and the accounts, without updates) has no JCL in the source.

4.2 The nightly batch

```
TRANBKP ─► POSTTRAN ─► INTCALC ─► TRANBKP ─► COMBTRAN ─► CREASTMT
backup,    daily       interest   backup     merge,      statements
empty      trans.      → SYSTRAN  again       reload       .PS and .HTML
KSDS       → TRANSACT; GDG                   TRANSACT
           rejects →
           DALYREJS GDG

TRANREPT: on demand; also submitted online by CORPT00C through the TD queue JOBS
```

4.3 Batch program interfaces

From each program's FILE-CONTROL (SELECT ... ASSIGN TO <DD name>) and CALL statements. "Random" and "sequential" are the ACCESS MODEs; the record layout comes from the copybook named.

Program	DD name	Organisation, access	Key	Record (copybook)
CBTRN01C	DALYTRAN	sequential	—	CVTRA06Y daily transaction
	CUSTFILE, XREFFILE, CARDFILE, ACCTFILE, TRANFILE	indexed, random	customer, card, card, account, transaction id	CVCUS01Y, CVACT03Y, CVACT02Y, CVACT01Y, CVTRA05Y
CBTRN02C	DALYTRAN in, DALYREJS out	sequential	—	CVTRA06Y (+ 80-byte reject trailer)
	TRANFILE, XREFFILE, ACCTFILE, TCATBALF	indexed, random	transaction, card, account, category-balance key	CVTRA05Y, CVACT03Y, CVACT01Y, CVTRA01Y
CBTRN03C	TRANFILE in, TRANREPT out, DATEPARM in	sequential	—	CVTRA05Y, report lines CVTRA07Y, dates
	CARDXREF, TRANTYPE, TRANCATG	indexed, random	card, type, type + category	CVACT03Y, CVTRA03Y, CVTRA04Y
CBACT04C	TCATBALF	indexed, sequential	account + type + category	CVTRA01Y
	XREFFILE (alternate key account), ACCTFILE, DISCGRP	indexed, random	card / account, account, group + type + category	CVACT03Y, CVACT01Y, CVTRA02Y
	TRANSACT out	sequential	—	CVTRA05Y; run date from the PARM (linkage)

Program	DD name	Organisation, access	Key	Record (copybook)
CBSTM03A	STMTFILE, HTMLFILE out	sequential	—	statement lines; file access via CALL 'CBSTM03B'
CBSTM03B	TRNXFILE, XREFFILE	indexed, sequential	card + transaction, card	COSTM01, CVACT03Y (passed by the caller)
	CUSTFILE, ACCTFILE	indexed, random	customer, account	CUSTREC, CVACT01Y
CBACT01C	ACCTFILE in; OUTFILE, ARRYFILE, VBRCFILE out	indexed sequential; sequential	account	CVACT01Y; three output formats; calls COBDATFT
CBACT02C, CBACT03C, CBCUS01C	CARDFILE, XREFFILE, CUSTFILE	indexed, sequential	card, card, customer	CVACT02Y, CVACT03Y, CVCUS01Y
CBEXPORT	five masters in; EXPFILE out	indexed, sequential	— / export sequence number	masters; CVEXPORT (500-byte multi-record)
CBIMPORT	EXPFILE in; CUSTOUT, ACCTOUT, XREFOUT, TRNXOUT, CARDOUT, ERROUT out	indexed sequential; sequential	export sequence number	CVEXPORT; the master layouts

All batch programs end abnormally through CALL 'CEE3ABD' (Language Environment) with a user abend code after an unexpected file status, displaying the status first (9910-DISPLAY-IO-STATUS, 9999-ABEND-PROGRAM).

4.4 Called routines

Routine	Language	Called by	Interface
CSUTLDTC	COBOL	COTRN02C, CORPT00C	Validate a date: linkage LS-DATE X(10), LS-DATE-FORMAT X(10), LS-RESULT X(80); calls the LE service CEEDAYS
CBSTM03B	COBOL	CBSTM03A	File access for the statements: one linkage area with DD name X(8), operation (O open, C close, R read, K read by key, W write, Z rewrite), return code X(2), key X(25) and key length, and a 1000-byte data area
COBDATFT	Assembler	CBACT01C	Convert a date between formats (input type, output type, date)

Routine	Language	Called by	Interface
MVSWAIT	Assembler	COBSWAIT	Wait for the interval given, with an interval-control timer
CEE3ABD, CEEDAYS	LE services	batch programs; CSUTLDTC	Abend with code; days since the Lilian epoch

5 Data

5.1 VSAM clusters

From the DEFINE CLUSTER statements in the loading jobs. Keys are given as length and offset. The full cluster name is AWS.M2.CARDDEMO.<name>.VSAM.KSDS; the table gives <name>.

Cluster	CICS file	Key (length, offset)	Record	Copybook	Content
ACCTDATA	ACCTDAT	11, 0 account id	300	CVACT01Y	Accounts: status, balances, limits, dates, group
CARDDATA	CARDDAT	16, 0 card number	150	CVACT02Y	Cards: account, CVV, embossed name, expiry, status
CUSTDATA	CUSTDAT	9, 0 customer id	500	CVCUS01Y	Customers: names, address, phones, SSN, FICO score
CARDXREF	CCXREF	16, 0 card number	50	CVACT03Y	Card → customer and account
TRANSACT	TRANSACT	16, 0 transaction id	350	CVTRA05Y	Posted transactions
USRSEC	USRSEC	8, 0 user id	80	CSUSR01Y	Users: names, password, type
TCATBALF	—	17, 0 account + type + category	50	CVTRA01Y	Balance per account and transaction category
DISCGRP	—	16, 0 group + type + category	50	CVTRA02Y	Interest rate per disclosure group and category
TRANTYPE	—	2, 0 type	60	CVTRA03Y	Transaction types
TRANCATG	—	6, 0 type + category	60	CVTRA04Y	Transaction categories
TRXFL	—	32, 0 card + transaction id	350	COSTM01	Work file of the statement job

5.2 Alternate indexes

The full name of an alternate index is AWS.M2.CARDDEMO.<name>.VSAM.AIX; the table gives <name>.

AIX	Base cluster	Key (length, offset)	Path (CICS file)	Used by
CARDDATA	cards	11, 16 account id (non-unique)	CARDAIX	defined (CSD, and as LIT-CARDFILENAME-ACCT-PATH in the card and account programs), but no EXEC CICS command uses it
CARDXREF	card cross reference	11, 25 account id (non-unique)	CXACAIX	COACTVWC, COACTUPC, COTRN02C, COBIL00C; INTCALC (DD XREFFIL1)
TRANSACT	transactions	26, 304 processing timestamp (non-unique)	—	defined by TRAN-FILE/TRANIDX

5.3 Sequential data sets and generation data groups

Data set (AWS.M2.CARDDEMO.)	Written by	Read by	Record
*.PS (ACCTDATA.PS, CARDDATA.PS, ...)	— (shipped sample data, EBCDIC)	loading jobs (REPRO)	the master layouts
DALYTRAN.PS	— (input of the day)	POSTTRAN, CBTRN01C	350, CVTRA06Y
DALYREJS (GDG)	POSTTRAN	—	430: daily transaction + reject reason (4 digits) and text (76)
TRANSACT.BKUP (GDG)	TRANBKP, TRANREPT	COMBTRAN, TRANREPT	350
SYSTRAN (GDG)	INTCALC	COMBTRAN	350 interest/fee transactions
TRANSACT.COMBINED (GDG)	COMBTRAN	reloaded into TRANSACT	350
TRXFL.SEQ	CREASTMT (SORT)	CREASTMT (REPRO into TRXFL.VSAM.KSDS)	350
STATEMNT.PS, STATEMNT.HTML	CREASTMT	—	80-byte statement lines, 100-byte HTML lines
TRANREPT (GDG)	TRANREPT	—	133-byte report lines (CVTRA07Y)
EXPORT.DATA	CBEXPORT	CBIMPORT	500, CVEXPORT; defined as a KSDS by the CBEXPORT job (both programs' RECORD KEY names a WORKING-STORAGE field: KD-12)

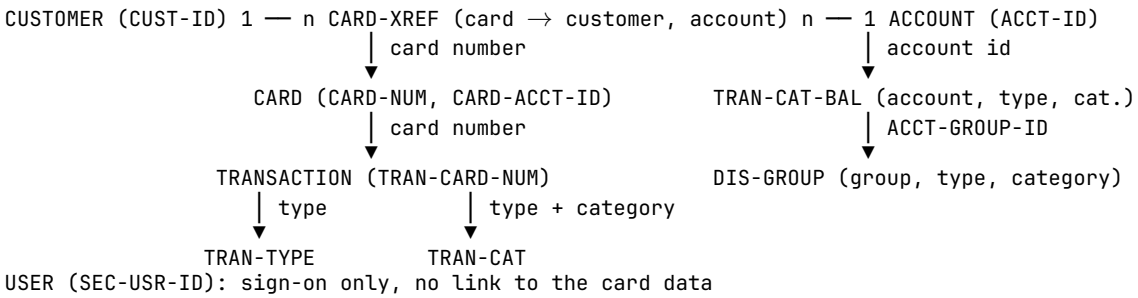
Data set (AWS.M2.CARDDEMO.)	Written by	Read by	Record
ACCTDATA.PSCOMP, .ARRYPS, .VBPS	READACCT	—	CBACT01C's demonstration formats

5.4 Record layouts

Copybook	Record (01 level)	Length	Key
CVACT01Y	ACCOUNT-RECORD	300	ACCT-ID 9(11)
CVACT02Y	CARD-RECORD	150	CARD-NUM X(16)
CVACT03Y	CARD-XREF-RECORD	50	XREF-CARD-NUM X(16); XREF-ACCT-ID alternate
CVCUS01Y	CUSTOMER-RECORD	500	CUST-ID 9(09)
CVTRA01Y	TRAN-CAT-BAL-RECORD	50	account 9(11) + type X(2) + category 9(4)
CVTRA02Y	DIS-GROUP-RECORD	50	group X(10) + type X(2) + category 9(4)
CVTRA03Y	TRAN-TYPE-RECORD	60	TRAN-TYPE X(2)
CVTRA04Y	TRAN-CAT-RECORD	60	type X(2) + category 9(4)
CVTRA05Y	TRAN-RECORD	350	TRAN-ID X(16)
CVTRA06Y	DALYTRAN-RECORD	350	DALYTRAN-ID X(16)
CVTRA07Y	report header, detail and total lines	112–133	—
COSTM01	TRNX-RECORD	350	card X(16) + transaction X(16)
CVEXPORT	EXPORT-RECORD	500	record type and sequence number; one layout per master record
CSUSR01Y	SEC-USER-DATA	80	SEC-USR-ID X(8)
COCOM01Y	CARDDEMO-COMMAREA	160	— (see the online subsystem)

Field-level layouts (every elementary item with its picture and offset) are in the requirements, chapter *Data dictionary*, which was derived from the same copybooks.

5.5 Relationships



6 External interfaces

Interface	Direction	Form
3270 terminal	in/out	BMS maps, 24 × 80; AID keys Enter, Clear, PF1–PF24
Daily transaction file	in	`DALYTRAN.PS`, 350-byte records
Report request	in	`DATEPARM` (start and end date) of `TRANREPT`; online, `CORPT00C` writes the JCL of a `TRANREPT` job to TD queue `JOBS`
Interest run date	in	PARM of `INTCALC` (`CBACT04C` linkage)
Statements, report, rejects	out	`STATEMNT.PS`/`.HTML`, `TRANREPT` GDG, `DALYREJS` GDG
Export file	out/in	`EXPORT.DATA`, 500-byte multi-record format (`CBEXPORT`, `CBIMPORT`)
Operator messages	out	`DISPLAY` to SYSOUT; abends via `CEE3ABD` with user codes

7 Source inventory

Kind	Folder	Count
COBOL programs, online	`cbl/CO*.cbl` (without `COBSWAIT`)	17
COBOL programs, batch and routines	`cbl/CB*`, `CSUTLDTC`, `COBSWAIT`	15
Assembler	`asm/`	2
Copybooks	`cpy/`	30
BMS mapsets	`bms/`	17
JCL members	`jcl/`	38
CICS resource definitions	`csd/CARDDEMO.CSD`	18 transactions, 8 files, 17 mapsets, 1 TD queue